
UNIT 10 MULTIVARIABLE REGRESSION MODELS

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10.0 OBJECTIVES

This Unit aims at equipping the learners with techniques of multiple regressions which come handy in analysing a number of situations where one is required to study impact of a number of independent variables (factors) on some particular dependent variable. After going through this unit, you will be able to:

- have a fairly good comprehension of technique of multiple regression analysis;
- appreciate the need for extending the basic idea of regression as discussed in Unit 8;
- apply those techniques for more realistic model formulation of explanation of economic phenomena; and
- spot and avoid pit falls which may arise (in quantitative analysis).

on accounts of auto-correlation, hetero-scedasticity and multi-co-linearity on the one hand and mis-specification of the model, (in the sense of including irrelevant as well as excluding relevant variables from the model) on the other.

10.1 INTRODUCTION

In Unit 8, you have studied basics of the Classical Linear Regression Model. You regressed dependent variable Y on independent variable X. In other

words, you tried to explain changes in Y in terms of the changes in X . You had hypothesised a linear relationship between Y and X of this type:

$$Y = \alpha + \beta X$$

Then, you went on to 'estimate' the two constants $\hat{\alpha}$ and $\hat{\beta}$. Once the estimates were ready, you had the estimated model or 'relationship' with you given by

$$\hat{Y} = \hat{\alpha} + \hat{\beta} x$$

In the present unit we are going to extend this type of analysis further to make it more 'realistic' and 'comprehensive'. We will do it in a number of steps: first of all, we shall introduce one more explanatory variable and re-examine the model. Next step will be to generalise the model to n -explanatory variables. At the next stage, we try to interpret the partial regression coefficients. After that, we will try to examine how large should the number ' n ' be – that is how many explanatory variables must be included in the model and what should be the statistical touch stone for arriving at such a decision. Then, we examine the conditions, or 'assumptions', which make these extensions and generalisations possible. We also examine the possible effects of violation of one or more assumptions. We shall, in particular, pay attention to problems of **multi-co-linearity**, **Hetero-scedasticity** and **auto-correlation**. Then, we take a look at another class of estimates, the Maximum Likelihood Estimators. Finally, we give a brief exposition to some uses of multiple regression analysis in economics.

10.2 REGRESSION MODEL WITH TWO EXPLANATORY VARIABLES

At the outset, please note one small change in notation: in section 10.1 – the introduction to the present unit, we have used the specification of model as

$$Y = \alpha + \beta X$$

However, now onwards, we will write the model in slightly different form:

$Y = \beta_0 + \beta_1 X_1$ This form helps us in presentation in the sense that as and when, we add more explanatory variables X_2, X_3 , etc. we simply indicate their coefficients with respective $\beta_1, \beta_2, \beta_3$, etc.

So the model that we consider in the present section has two explanatory variables X_1 , and X_2 . The non-stochastic specification will be:

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \quad (10.1)$$

The same model in stochastic form will have 'error terms' included:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \mu \quad (10.2)$$

$$= E(Y) + \mu \quad (10.3)$$

We can use subscripts ' t ' with Y_0, X_1, X_2 and μ to denote ' t^{th} ' observation. Thus, the above equations can be written as

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t}$$

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \mu_t \text{ and}$$

$$Y_t = E(Y_t) + \mu_t \text{ respectively}$$

In the relationships above, the component $(\beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t})$ is systematic or deterministic component, which is equal to mean value $E(Y_t)$ — or a point on the regression line.

The component μ_t is non-systematic random component determined by factors other than X_1 and X_2 .

The model specified by equations 10.1 and 10.2 is a linear regression model which is linear in parameters.

10.2.1 Estimation of Parameters: The Ordinary Least Squares Approach

The Ordinary Least Squares Approach

We collect a sample of observations on Y_1 , X_1 and X_2 and write down sample regression function

$$Y_t = b_0 + b_1 X_{1t} + b_2 X_{2t} + e_t \quad (10.4)$$

Note that b_0 , b_1 , b_2 have replaced the corresponding population parameters β_1 , β_2 , β_3 here. The random component μ_t replaced by e_t or the sample error term.

So, simply speaking:

b_0 = the estimator for β_0

b_1 = the estimator for β_1

b_2 = the estimator for β_2

We know from the Unit 8 that the *principle of ordinary least squares* selects those values for unknowns (b_0, b_1, b_2) such that residual sum of squares (RSS) $= \sum e_t^2$ is minimum. We can develop this idea by rewriting equation 10.4 as

$$e_t = Y_t - b_0 - b_1 X_{1t} - b_2 X_{2t} \quad (10.5)$$

Square on both sides and sum up to get RSS:

$$\sum e_t^2 = \sum [Y_t - b_0 - b_1 X_{1t} - b_2 X_{2t}]^2 \quad (10.6)$$

Differentiating the (10.6) w.r.t b_0 , b_1 , b_2 and equating to zero gives us the three normal equations:

$$\bar{Y}_t = b_0 + b_1 \bar{X}_{1t} + b_2 \bar{X}_{2t} \quad (10.7)$$

$$\sum Y_t X_{1t} = b_0 \sum X_{1t} + b_1 \sum X_{1t}^2 + b_2 \sum X_{1t} X_{2t} \quad (10.8)$$

$$\sum Y_t X_{2t} = b_0 \sum X_{2t} + b_1 \sum X_{1t} X_{2t} + b_2 \sum X_{2t}^2 \quad (10.9)$$

These three equations give us the following expressions for b_0 , b_1 , and b_2 respectively:

$$b_0 = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2 \quad (10.10)$$

$$b_1 = \frac{(\sum y_t x_{1t})(\sum x_{2t}^2) - (\sum y_t x_{2t})(\sum x_{1t} x_{2t})}{(\sum x_{1t}^2)(\sum x_{2t}^2) - (\sum x_{1t} x_{2t})^2} \quad (10.11)$$

and

$$b_2 = \frac{(\sum y_t x_{2t})(\sum x_{1t}^2) - (\sum y_t x_{1t})(\sum x_{1t} x_{2t})}{(\sum x_{1t}^2)(\sum x_{2t}^2) - (\sum x_{1t} x_{2t})^2} \quad (10.12)$$

The lower case letters denote, as usual, the deviations from the respective means. Thus $y_t = (y_t - \bar{Y})$, $x_{1t} = (X_{1t} - \bar{X}_1)$ and $x_{2t} = (X_{2t} - \bar{X}_2)$..

10.2.2 Variance and Standard Errors

The variances of OLS estimators given in 10.10, 10.11 and 10.12 above are given in terms of means and deviations of X_1 and X_2 and the variance of population error terms μ_t . Thus,

$$Var(b_0) = \left(\frac{1}{n} + \frac{\bar{X}_1^2 \sum x_{2t}^2 + \bar{X}_2^2 \sum x_{1t}^2 - 2 \bar{X}_1 \bar{X}_2 \sum x_{1t} x_{2t}}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \right) \sigma^2 \quad (10.13)$$

$$\text{Standard Error of } b_0 = \sqrt{Var(b_0)} \quad (10.14)$$

$$Var(b_1) = \frac{\sum x_{2t}^2}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \cdot \sigma^2 \quad (10.15)$$

$$SE(b_1) = \sqrt{Var(b_1)} \quad (10.16)$$

and

$$Var(b_2) = \frac{\sum x_{1t}^2}{\sum x_{1t}^2 \sum x_{2t}^2 - (\sum x_{1t} x_{2t})^2} \cdot \sigma^2 \quad (10.17)$$

$$SE(b_2) = \sqrt{Var(b_2)} \quad (10.18)$$

When σ^2 is unknown and its unbiased OLS estimator is obtained, we find that

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{n-3} \quad (10.19)$$

The denominator of 10.19 shows the degrees of freedom. In a sample of size 'n' we exhaust 3 degrees of freedom in estimating b_0 , b_1 , and b_2 . Note that this reasoning is quite general, in the sense that if out of a sample of size 'n', we

estimate 'k' parameters, remaining degrees of freedom will be $n - k$. We call $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$, the standard error of estimate/regression.

The residual sum of squares in 10.19 is $\sum e_t^2 = \sum (Y_t - \hat{Y}_t)^2$. This expression is equivalent to the following.

$$\sum e_t^2 = \sum y_t^2 - b_1 \sum y_t x_{1t} - b_2 \sum y_t x_{2t} \quad (10.20)$$

10.3 INTERPRETATION OF REGRESSION COEFFICIENTS

Mathematically, b_1 and b_2 represent the partial slopes of regression plan with respect to X_1 and X_2 respectively. In other words, b_1 shows the rate of change in Y as X_1 alone undergoes a unit change, keeping all other thing, constant. Similarly, b_2 represents rate of change of Y as X_2 alone changes by a unit while other things are held constant.

Note: This interpretation is quite similar to what you have been doing in economic theory. Recall the law of demand: quantity demanded of a commodity varied inversely with its price, 'ceteris paribus' that is when other things were kept 'constant'. What were those other things? Those were prices of complements and substitutes, income of the consumer, and tastes/likings/disliking etc.

10.3.1 Goodness of Fit: Multiple Coefficient of Determination: R^2

Recall that in case of a single independent variable, r^2 measured goodness of fit of the fitted sample regression line. It gave you the percentage total variation in dependent variable Y which has been explained by the single explanatory variable X . Correspondingly, when we have two explanatory variables X_1 and X_2 , we would like to know the proportion of total variation in $Y = \sum y_t^2$ explained by X_1 and X_2 jointly. This information is conveyed by R^2 — the multiple coefficient of determination. Thus,

$$R^2 = \frac{ESS}{TSS}$$

When ESS is explained sum of squares and TSS is total sum of squares. You are familiar with the relationship between TSS, ESS and RSS from Unit 8.

You know that $ESS = b_1 \sum y_t x_{1t} + b_2 \sum y_t x_{2t}$

and $RSS = \sum y_t^2 - b_1 \sum y_t x_{1t} - b_2 \sum y_t x_{2t}$

Therefore R^2 can be computed as

$$R^2 = \frac{b_1 \sum y_t x_{1t} + b_2 \sum y_t x_{2t}}{\sum y_t^2} \quad (10.21)$$

You can get the quantities in the above formula from your computation sheet for the normal equations. The R^2 lies between 0 and 1. Closer it is to one;

better is the fit – implying estimated regression line is capable of explaining greater proportions of variation in Y. The positive square root of R^2 is called coefficient of multiple correlation.

10.3.2 Analysis of Variance (ANOVA)

We know the relationship:

$$TSS = ESS + RSS$$

This is equivalent to saying:

$$\sum y_i^2 = b_1 \sum y_i x_{1i} + b_2 \sum y_i x_{2i} + \sum e_i^2$$

$$(TSS) = (ESS) + (RSS)$$

A study of the components of TSS is called analysis of variance in the context of regression. You know that every sum of squares has some degrees of freedom (df) associated with it. In our above 2-explanatory variable case, the degrees of freedom will be

$$TSS = n-1$$

$$RSS = n-3$$

$$ESS = 2$$

We can put this information in form of a table:

Table 10.1: ANOVA Table for 2-Explanatory Variable Regression

Source of Variation	Sum of Squares (SS)	df	Mean S.S = $\frac{SS}{df}$
Due to regression (ESS)	$b_1 \sum y_i x_{1i} + b_2 \sum y_i x_{2i}$	2	$\frac{ESS}{2}$
Due to residuals	$\sum e_i^2$	n-3	$\frac{\sum e_i^2}{n-3}$
TSS	$\sum y_i^2$	n-1	

One may be interested in testing a null hypothesis $H_0: B_1 = B_2 = 0$. In such a case we find that

$$\frac{ESS / df}{RSS / df}$$

that is ratio of the variance explained by X_1 and X_2 to unexplained

variance follows F distribution with 2 and n-3 degrees of freedom. In general, if regression equation estimates 'k' parameters including intercept, then F ratio has (k-1) df in numerator and (n-k) df in the denominator.

Interpretation: Larger the variance explained by fitted regression line, larger the numerator will be in relation to denominator. Thus a larger F value is evidence **against** the 'truthfulness' of $H_0: B_1 = B_2 = 0$. That is, F values larger than one will indicate that hypothesis of both the variables X_1 and X_2 having no effect on Y cannot sustain.

We can also express F values in terms of R^2 .

$$F = \frac{R^2 / (k - 1)}{(1 - R^2) / (n - k)}$$

Check Your Progress 1

1) What is multiple regression? Explain with an example.

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2) State the assumptions of classical linear multiple regression model.

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3) What do you mean by linearity of regression model? Explain?

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4) How do you interpret coefficients of multiple regression model?

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10.4 INCLUSION AND EXCLUSION OF EXPLANATORY VARIABLES

The Adjusted $R^2 : \bar{R}^2$

It has been noted that as we add more and more explanatory variables X_s , the explained sum of squares keeps on rising. Thus, R^2 goes on increasing as we increase X_s , the explained sum of squares keeps on rising. Thus, R^2 goes on increasing as we increase X_s . But, notice that adding each additional variable 'eats up' one degree of freedom and our definition of R^2 makes no allowance for loss of degrees of freedom. Hence, the thinking that you can improve the goodness of fit by suitably increasing the number of variables may not be justified — We know that TSS always has $(n-1)$ degrees of freedom. Therefore, comparing two regression models with same dependent variable but differencing number of independent variables will not be justified. Therefore, we must adjust our measure of goodness of fit for degrees of freedom. This measure is called adjusted R^2 , denoted by $\bar{R}^2$. It can be derived from R^2 as follows:

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)} \quad (10.22)$$

Therefore, it is recommended that, one must include new variables only if (upon inclusion) $\bar{R}^2$ increases and not otherwise. A general guide is provided by 't' statistic, if absolute value of the co-efficient of added variable is greater than one, retain it (Note that 't' value is calculated under the hypothesis that population value of that co-efficient is zero).

10.5 GENERALISATION TO N-EXPLANATORY VARIABLES

In general, our regression model may have a large number of independent variables. Each of those variables can, on priority grounds, be expected to have some influence over the 'dependent' or 'explained' variable. Consider a very simple example. What can be determinants of demand for potatoes in a vegetables market? One obvious choice will be the price of potatoes. What else can affect the quantity demanded? Could it be availability of vegetables which can be paired off with potatoes? In that case, prior of a large number of vegetables which are cooked along with potatoes will become 'relevant explanatory variables'. You cannot ignore income of the community that patronizes the particular market. Needless to say, the dietary preferences of members of the households can also affect the demand and so on. In the next Unit, we shall discuss techniques which help us restrict the analysis to a selected 'few variables, though theoretic considerations may find a huge number of them to be 'useful' and 'powerful' determinants. In fact, in economic theory, we usually append the phrase **Ceteris paribus**, with many a statements. This phrase means keeping all other things constant. That means, we may focus on impact of only a few selected variables on the dependent variable while assuming that all other variables remain 'unchanged' during the period of analysis. This assumption may not hold so, we have to juggle the need to include more and more variables in our model with the 'gains' in

explanatory power of the model. We have developed, in previous section (10.4) a working touchstone for inclusion of more variables in terms of improvement in $\overline{R^2}$ and have tried to give it 'practical' shape in form of magnitude of 't' values of the relevant slope parameters.

With these considerations in mind we can generalise the linear regression model as follows:

We hypothesize that in population, the dependent variable Y depends upon n explanatory variables, $X_1, X_2, \dots, X_n$. We also assume that the relationship is linear in parameters. Three more assumptions are made and they have very significant bearing on the analysis. These are:

- a) Absence of Multi-co-linearity;
- b) Absence of Hetero-scedasticity; and
- c) Absence of Autocorrelation

We will discuss the complications, which arise because of violations of these assumptions in section 10.6, 10.7 and 10.8 respectively. So our Classical Linear General Regression Model is :

$$Y_t = \beta_0 + \beta_1 X_{1t} + \dots + \beta_n X_{nt}$$

or

where it is understood that Y and each X will have a large numbers of values $t=1, \dots, N$, forming (n+1) 'tuples' ($Y_{11}, X_{11}, X_{21}, \dots, X_{n1}$).

We can simply write

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \mu \quad (10.23)$$

in Matrix equation form, we get

$$Y = X\beta + U \quad (10.24)$$

Where $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$ and $U = \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix}$

also $X = \begin{bmatrix} X_{11}, X_{21}, X_{31}, \dots, X_{k1} \\ X_{12}, X_{22}, X_{32}, \dots, X_{k2} \\ \vdots \\ X_{1n}, X_{2n}, X_{3n}, \dots, X_{kn} \end{bmatrix}$

We assume that (1) expected values of error terms are equal to zero; that is $E(u_i) = 0$ for all 'i'. In matrix notation

$$E(u) = \begin{bmatrix} E(u_1) \\ \vdots \\ E(u_n) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0$$

- 2) The error terms are not correlated with one another and they all have same variance for σ^2 all sets values of the variables X. That is,

$$E(u_i u_j) = 0; \quad \forall i \neq j \text{ and}$$

$$E(u_i^2) = \sigma^2 \forall i$$

in matrix notation:

$$E[UU'] = \begin{bmatrix} E(u_1^2), E(u_1 u_2) \dots \dots \dots E(u_1 u_n) \\ E(u_1 u_2), E(u_2^2) \dots \dots \dots E(u_2 u_n) \\ \vdots \\ E(u_1 u_n), E(u_2 u_n) \dots \dots \dots E(u_n^2) \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 \\ 0 & \sigma^2 & \sigma & 0 \\ 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & \sigma^2 \end{bmatrix} = \sigma^2 I_n$$

- 3) Number $X_{1i} \dots \dots \dots X_{ki}$ are all real numbers and are free of random errors or disturbances.
- 4) The matrix X has been linearly independent columns. It implies that number of observations exceeds number of co-efficient to be estimated. It also implies there exists no exact linear relationships between any of the X variables.

Note: Assumptions that $E(U_i U_j) = 0$ means that error terms are not correlated. The implication of **diagonal** terms in matrix $E(UU')$ being all equal to σ^2 is that all error terms have same variance, σ^2 . This is also called assumption of homo-scedasticity. The last assumption implies absence of multi-co-linearity. We can write the regression relation for the sample as:

$$e = Y - Xb$$

where e , Y , X and b are appropriate matrices.

Sum of squared residuals will be

$$\begin{aligned} \phi &= \sum e_i^2 = \sum (Y_i - b_1 X_{1i} \dots \dots \dots + b_k X_{ki})^2 \\ &= e'e = [Y - Xb]' [Y - Xb] \\ &= Y'Y - 2b'X'Y + b'X'Xb \end{aligned} \quad (10.25)$$

Note: $b'X'Y$ is scalar and is therefore equal to its transpose $Y'Xb$.

by equating 1st order partials of ϕ w.r.t each b_i , to zero, we get k normal equations. This set of equations in matrix form is: